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From Motion to Behavior: Understanding Aggressive Driving Using Smartphone Sensors

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ABSTRACT

Smartphone motion sensors offer a scalable framework for monitoring driving behavior, yet many existing machine learning approaches lack interpretability. In this project, we quantify the association between frequency-domain motion features and aggressive driving behavior using smartphone accelerometer and gyroscope data. Labeled driving events were segmented into fixed-length windows, and Fast Fourier Transform (FFT) was applied to extract spectral energy and spectral centroid features. These window-level predictors were analyzed using a mixed-effects logistic regression model to account for within-event correlation. Results indicate that rotational dynamics, particularly gyroscope-based energy and frequency characteristics, are strongly associated with aggressive driving behavior. This framework demonstrates that interpretable frequency-domain modeling can effectively translate raw sensor data into meaningful driving risk indicators.

INTRODUCTION

Road traffic injuries remain one of the leading causes of preventable morbidity and mortality worldwide. According to global transportation safety reports, risky driving behavior is strongly associated with increased crash probability and greater injury severity. Aggressive driving behavior not only endangers drivers and passengers but also imposes significant economic burdens on healthcare systems, insurance providers, and transportation industries (World Health Organization 2023; Blincoe et al. 2015).

With the rapid proliferation of smartphones equipped with motion sensors, studies have shown that smartphone sensor data can be used to detect aggressive driving behavior with high accuracy by applying machine learning algorithms to time-series sensor signals (Ferreira Júnior et al. 2017). These approaches demonstrate that passive mobile sensing provides a scalable and cost-effective framework for monitoring driving behavior.

However, many machine learning approaches operate as black-box classifiers. While they may achieve strong predictive performance, they often do not provide interpretable estimates of how individual sensors contribute to aggressive driving risk. Interpretable estimates are essential for understanding the underlying motion dynamics associated with risky behavior, facilitating regulatory or insurance applications, and providing actionable feedback to drivers. Therefore, beyond classification accuracy, our objective is to quantify how motion dynamics captured by smartphone sensors are associated with the probability of aggressive driving behavior.

In this project, we focus on two primary smartphone sensors: the accelerometer, which measures linear acceleration, and the gyroscope, which measures rotational motion. We applied the Fast Fourier Transform (FFT) technique to the sensor time-series data to extract frequency-domain features that characterize motion intensity and oscillatory patterns. We then built a mixed-effects logistic regression model to evaluate the association between

these sensor-derived features and the probability of aggressive driving behavior and quantified the contribution of each feature on driving risk from the estimated coefficients.

DATA DESCRIPTION

The dataset in this project consists of smartphone motion sensor measurements collected during controlled driving experiments designed to capture aggressive and non-aggressive driving behaviors (Ferreira Júnior 2017).

The experiments were conducted across four driving trips, each lasting approximately thirteen minutes, using a 2011 Honda Civic under sunny weather conditions on dry, paved asphalt roads. During each trip, predefined driving maneuvers were executed and recorded with the start and end times of each driving event for labeling purposes.

The smartphone (Motorola XT1058, Android 5.1) continuously recorded high-frequency motion signals from embedded sensors, including tri-axial acceleration, linear acceleration, angular velocity (gyroscope), and magnetometer measurements. The sampling rate ranged between approximately 50 Hz and 200 Hz depending on the sensor.

Driving events were labeled by aligning sensor timestamps with the recorded video. Each event corresponded to a specific maneuver and was categorized into one of seven types: aggressive braking, aggressive acceleration, aggressive left turn, aggressive right turn, aggressive left lane change, aggressive right lane change, or non-aggressive driving. A total of 69 labeled events were collected across the four trips.

In this project, accelerometer and gyroscope signals were used for analysis, and the seven maneuver types were grouped into a binary outcome (aggressive vs. non-aggressive) to focus on distinguishing overall aggressive driving behavior at a broader behavioral level.

DATA CLEANING AND FEATURE ENGINEERING

FREQUENCY-DOMAIN TRANSFORMATION (FFT)

Smartphone accelerometer and gyroscope signals are continuous time-series measurements that describe how a vehicle moves over time. Extracting relevant and informative features that accurately characterize motion dynamics can be challenging when relying solely on time-domain representations. Time-domain signals capture magnitude changes but may not clearly distinguish between smooth and abrupt movement patterns.

To better characterize motion dynamics, we represented each sensor signal in the frequency domain using the Fast Fourier Transform (FFT). The FFT decomposes discrete time series data into a set of sinusoidal components at different frequencies. Rather than describing motion solely by its amplitude at each time point, the frequency-domain representation characterizes how signal energy is distributed across oscillation rates (Oppenheim and Schaffer 2014).

This representation is suitable for aggressive driving analysis because abrupt maneuvers introduce stronger high-frequency components into acceleration and rotational signals. In contrast, smooth driving behavior concentrates energy at lower frequencies, reflecting gradual and stable motion patterns.

EVENT ALIGNMENT AND SEGMENTATION

We first aligned the continuous accelerometer and gyroscope recordings with the labeled event intervals. For each annotated driving maneuver, sensor observations whose timestamps fell within the corresponding start and end times were extracted, producing event-specific time-series segments containing tri-axial acceleration and angular velocity measurements.

To standardize the analytical unit, we segmented each event-level time series into overlapping subsegments using a sliding-window procedure. Specifically, we generated windows of 128 consecutive samples and advanced the starting position by 10 samples to create overlapping windows. The window length of 128 ensures consistent input size for frequency-domain transformation and provides adequate resolution for capturing short-duration motion fluctuations. The overlapping design increases the number of observations while preserving temporal continuity within each event.

FEATURE EXTRACTION AND DATASET CONSTRUCTION

For each 128-sample window, we computed the discrete Fourier transform separately for each axis of the accelerometer and gyroscope. From the resulting power spectrum, we derived two summary features per axis: spectral energy and spectral centroid. Spectral energy was calculated as the total power across non-zero frequencies, representing the overall intensity of motion within the window. Spectral centroid was calculated as the power-weighted mean frequency, representing the dominant oscillation frequency of the motion signal.

To reduce dimensionality, we aggregated axis-level measures into sensor-level features. Spectral energies across the three axes were summed to obtain total accelerometer energy and total gyroscope energy. For the centroid measure, we selected the centroid corresponding to the axis with the highest spectral energy within each window to represent the dominant direction of motion. Because energy measures are typically right-skewed, logarithmic transformations were applied to stabilize variance (Kutner et al. 2013).

The final analytical dataset contained 292 window-level observations. Each observation corresponded to a 128-sample subsegment derived from a labeled driving event and included the binary aggressive-driving indicator, an event identifier to account for within-event correlation, and four frequency-domain predictors: total accelerometer spectral energy, total gyroscope spectral energy, accelerometer spectral centroid, and gyroscope spectral centroid. This window-level structure formed the basis for the subsequent mixed-effects logistic regression analysis.

ANALYSIS

The objective of the analysis was to evaluate the association between frequency-domain sensor features and the probability of aggressive driving behavior. Because multiple window-level observations were derived from the same driving event, the data exhibit a hierarchical structure in which windows are nested within events. Observations originating from the same event are expected to be correlated due to shared driving context and maneuver characteristics.

To account for this within-event correlation, we implemented a mixed-effects logistic regression model with a random intercept for each driving event using PROC GLIMMIX (METHOD=LAPLACE) in SAS. Let Y_{ij} denote the binary indicator of aggressive driving for window j within event i . The model was specified as:

$$\text{logit}(P(Y_{ij} = 1)) = \beta_0 + \beta_1 E_{1ij} + \beta_2 E_{2ij} + \beta_3 C_{1ij} + \beta_4 C_{2ij} + u_i$$

where:

- E_{1ij} = log-transformed total accelerometer spectral energy
- E_{2ij} = log-transformed total gyroscope spectral energy
- C_{1ij} = log-transformed accelerometer spectral centroid
- C_{2ij} = log-transformed gyroscope spectral centroid
- $u_i \sim N(0, \sigma^2)$ = random intercept for event i

Model parameters were estimated using maximum likelihood with the Laplace approximation. The Laplace method provides an accurate likelihood-based approximation for generalized linear mixed-effects models with binary outcomes and random effects.

Estimated coefficients were interpreted on the log-odds scale. Each coefficient represents the expected change in the log-odds of aggressive driving associated with a one-unit increase in the corresponding predictor, holding other variables constant.

Model discrimination was assessed using the area under the receiver operating characteristic curve (ROC). Predicted probabilities were obtained from the fitted mixed-effects model, and the ROC curve was constructed to evaluate the model's ability to distinguish between aggressive and non-aggressive windows.

RESULTS

Among the four frequency-domain predictors, three showed statistically significant associations with aggressive driving behavior.

Total gyroscope spectral energy was positively associated with aggressive driving ($\hat{\beta}_2 = 23.91$, $p = 0.0033$). This indicates that higher rotational motion intensity substantially increases the log-odds of aggressive driving. Rotational dynamics appear to be a strong indicator of aggressive maneuvers such as sharp turns or abrupt lane changes.

Accelerometer spectral centroid was negatively associated with aggressive driving ($\hat{\beta}_3 = -19.27$, $p = 0.0397$). This suggests that windows dominated by higher-frequency acceleration components are less likely to be classified as aggressive after accounting for other features. In contrast, the gyroscope spectral centroid showed a strong positive association ($\hat{\beta}_4 = 47.26$, $p = 0.0001$), indicating that higher-frequency rotational oscillations are strongly linked to aggressive behavior.

Total accelerometer spectral energy showed a positive but not statistically significant association ($\hat{\beta}_1 = 7.79$, $p = 0.0888$), suggesting that linear acceleration intensity alone may be less discriminative than rotational dynamics when both sensors are considered jointly.

The estimated variance of the random intercept was substantial ($\hat{\sigma}_u^2 = 10507$), indicating considerable variability in baseline aggressive-driving probability across events. This pronounced clustering suggests that windows derived from the same labeled event exhibited highly consistent motion patterns, implying that aggressive driving reflects sustained dynamic structure rather than transient signal noise.

The model demonstrated strong discriminative ability. The area under the receiver operating characteristic curve (AUC) was 0.973 (95% CI: 0.945–1.000), indicating a high ability to distinguish aggressive from non-aggressive windows.

The results indicated that frequency-domain features derived from smartphone motion sensors were strongly associated with aggressive driving behavior. Gyroscope-based measures showed the strongest effects, suggesting that abrupt directional changes and rapid angular movements are key signatures of aggressive maneuvers.

DISCUSSION

The positive association between gyroscope spectral energy and aggressive driving implies that sharper and more forceful rotational movements increase driving risk. Higher gyroscope spectral centroid values further indicate that aggressive behavior is characterized by faster oscillatory motion, reflecting sudden steering inputs and rapid directional adjustments. In contrast, linear acceleration measures showed comparatively weaker effects, suggesting that rotational instability may be a more distinctive indicator of aggressive behavior than longitudinal force alone.

These findings suggest that aggressive driving is defined not only by movement magnitude but also by the dynamic structure of motion patterns. In real-world applications, such interpretable frequency-based indicators could be integrated into smartphone-based driver monitoring systems to identify high-risk maneuvers in real time, support usage-based insurance programs, or provide feedback to drivers aimed at improving road safety.

CONCLUSION

This project demonstrated that frequency-domain features derived from smartphone accelerometer and gyroscope signals can effectively characterize aggressive driving behavior. By combining Fast Fourier Transform with mixed-effects logistic regression, we quantified how motion intensity and oscillatory dynamics are associated with driving risk. The framework provides a transparent and statistically grounded approach for translating raw sensor data into interpretable risk indicators.

FUTURE WORK

Future research should evaluate the proposed framework using larger and more diverse driving datasets to assess robustness across different drivers, road environments, vehicle types, and traffic conditions. In particular, future studies could examine performance under varying environmental conditions such as wet or slippery roads, nighttime driving, heavy traffic, and uneven terrain, where motion patterns may differ substantially from controlled daytime settings. Validation under real-world variability is essential to determine generalizability beyond controlled experimental conditions and to ensure reliable deployment in practical driver-monitoring and safety applications.

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